Multi-Stakeholder Explainable Job Recommendation Systems

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I. Context

In recent years, Explainable Artificial Intelligence (XAI) has gained considerable traction, becoming an essential part of Artificial Intelligence (AI) research and development [1], [2]. The need for transparency and interpretability in AI systems has led to advancements in XAI, benefitting developers, data scientists, and AI experts [3], [4].

Nevertheless, a substantial gap persists in converting these complex technical explanations into easily understandable insights for the general public.

This disparity is especially noticeable within the domain of multi-stakeholder job recommender systems, where the objective is to connect job seekers with employers, demanding explanations that are accessible to a broad and varied audience [5], [6].

The field of XAI has evolved to address the need for transparency and interpretability in machine learning models [1]. Various techniques have been proposed, including rule-based explanations [3], feature importance analysis [4] and model-specific interpretability approaches [7]. However, these methods, while beneficial for technical experts, often fall short in providing intuitive explanations for non-technical stakeholders.

In parallel, multi-stakeholder job recommender systems have emerged to facilitate the matchmaking process between job seekers and employers [5]. These systems require recommendations that are not only comprehensible but also actionable for job seekers, employers, and regulatory bodies [6]. Despite advancements, existing approaches struggle to offer personalized, transparent, and user-friendly explanations for these diverse stakeholders.

Another prior study has investigated the challenges of multi-stakeholder job recommender system explanations [8], where a co-design study was performed to enhance the explanations provided by such systems, contributing valuable insights into the practical aspects of bridging the gap between XAI and multi-stakeholder job recommender systems [8]. Exploring three explanation techniques: text-based explanation, graph-based explanation, and feature attribution, where

1) Candidates prefer short clearly-structured text containing the most critical information [8].
2) Recruiters also prefer text-based explanation, but allowing it to be longer containing more detailed information [8].
3) Company Representatives prefer graph-based explanations, allowing them to get a comprehensive overview of more complex explanations [8].

Furthermore, the study discussed future work on performing research on other explanation techniques such as hybrid explanations, attention mechanisms, saliency, and post-hoc methods [8].

In light of these challenges, this research paper seeks to explore novel methodologies that bridge the gap between XAI and multi-stakeholder job recommender systems. By drawing upon insights from both fields, we aim to develop effective approaches for explaining job recommendations to a broad audience with varying levels of technical expertise, ultimately enhancing the utility and transparency of such systems.

A. Research Questions

To guide our study, we formulate the following research questions:

1) What are user-centric interfaces that facilitate intuitive exploration and understanding of job recommendations?
2) Given a job recommender system implemented with machine learning, which explanation types can be applied to explain the outcome of this system?
3) Given a mock-up job recommender system, which explanation types can be applied to explain the outcome of this system?
4) How can we evaluate the explainability of job recommendations?

These research questions will serve as the foundation for our exploration and the subsequent development of methodologies to address the challenges at hand.

II. Related Work

The field of XAI has seen much more attention in recent years, including explanations for Recommender Systems. However, this is a very broad topic, with there being different use-cases and ethical reasons to have in mind while generating explanations.

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A. Evaluating Explanations

Considering this wide-ranging topic, [9] has proposed 7 possible aims for an explanation. Table I defines these aims, encompassing a comprehensive perspective that assesses explanations from various angles. This evaluation encompasses factors such as users' confidence in the system, indicated by the measure of Trust, as well as ensuring user engagement with recommended products through the incorporation of Persuasive explanations. In accordance with these aims, [10] has assessed various explanation types and formulated some evaluation guidelines. For efficiency, they measure the time it takes a user to evaluate an item, while effectiveness involves comparing the user experience with and without an explanation accompanying the recommendation.

Nevertheless, it's crucial to recognize that within the context of MSRSs, certain aims hold greater significance for specific stakeholders than others and should be looked at from a different viewpoint. Given that these explanations impact various stakeholders, it is imperative to not only assess their satisfaction with one party but also ensure that they do not have detrimental effects on others. As an example, in [11] the authors analyzed several types of explanations in the context of MSRS. Contrastive explanations were discussed, with this type of explanation one wants to answer "Why P and not Q, for example, an XAI image classifier should be able to explain "Why did you classify that photo as a cat, not a dog?". In the context of MSRS, this approach is unsuitable, as it may indirectly reveal one of the stakeholder's preferences. For example, in a marketplace with consumers and providers, a contrastive explanation concerning the consumer's question regarding the lower ranking of the Samsung Galaxy S21 compared to the Apple Phone XS, despite the consumer's liking them both equally, would have no explanation to put forward besides the provider's preferences. No explanation seems both transparent and satisfactory to both parties. It seems that for systems that have multiple stakeholders, a different objective should be taken into account and prioritized, Fairness to all parties.

B. Fairness in MSRS

Many studies have addressed the fairness objective [12], [13], and [14]. However, these studies predominantly concentrate on the recommendation engine itself, and there is a limited body of research regarding fair explanations. As [8] states, previous work on explainability in MSRS, and more specifically Job Recommender Systems(JRSs), is very limited and seemingly an afterthought, with them mostly failing to include tailored explanations to all stakeholders. The task of a fair explanation is very difficult to generalize, with structured requirements being necessary from each specific stakeholder. This paper aims to ensure this fairness principle while exploring and creating various types of explanations.

C. Explanation Types

Throughout the years, many explanation types have been put forth [10], [15], and [16]. These explanations have primarily been tailored to environments characterized by a single stakeholder, typically the user of a platform that provides recommendations for media, such as music or films. For example, [17] found in their study that a histogram with a grouping interface performed best (Figure 1). In the context of JRSs, this explanation approach may not be practically applicable, as neither stakeholder typically provides ratings for the product, which could be either a job post or an employee profile. However, under the assumption of data availability, it would be conceivable to replace rating values with alternative indicators that signify the worthiness of a recommendation. Some other effective explanation types include various styles of textual explanations, such as those grounded in demographic information [18], utility-driven explanations [19], or conversational-style explanations [20]. In a study conducted by Kleinerman et al. [21], an investigation into explainability in online dating revealed that explanations incorporating mention of both parties involved tend to outperform conventional, one-sided explanations in high-stakes environments, such as job recruitment. The concept of considering both parties can be effectively applied to the aforementioned textual explanation styles, rendering them more suitable for the context of JRSs.

This approach of taking explanations that performed well in regular RSs and converting them to explanations of JRSs has not been explored too much in the field and will be investigated further in this paper.

Aim	Definition
Transparency	Explain how the system works
Scrutability	Allow users to tell the system it is wrong
Trust	Increase users' confidence in the system
Efficiency	Help users make decisions faster
Satisfaction	Increase the ease of use or enjoyment
Effectiveness	Help users make good decisions
Persuasiveness	Convince users to try or buy

TABLE I: Seven Possible aims of Explanations

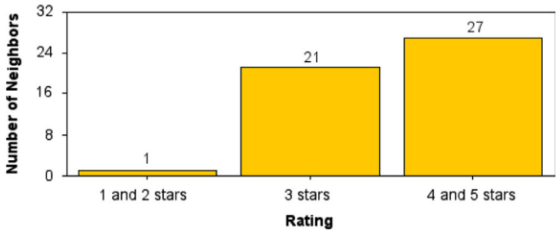


Fig. 1: Histogram with a grouping interface, which performed best in [17].The x-axis displays similar users' ratings, and y shows the number of similar users that gave that rating.

III. Concepts and Approach

A. Explanation Types

1) Textual:

a) Counterfactual: Counterfactual explanations provide an insightful "what-if" scenario by suggesting an alternative

course of action [11]. In the context of a recommendation system, a counterfactual explanation might say, "I recommended product A because you liked product B the most. If your preference had been different, I would have suggested product C instead."

b) Contrastive: Contrastive explanations highlight the difference or discrepancy between two choices, helping users understand why one option was chosen over another [11]. For instance, in job recommendations, a contrastive explanation might explain why Job X was recommended instead of Job Y, emphasizing factors such as proximity or qualifications.

c) Conversational: These explanations engage users in a conversation [20], allowing them to ask questions, seek additional information, or request further insights regarding the recommended job. Conversational explanations promote user engagement, offer personalized responses, and address specific queries.

d) Demographic: Demographic explanations consider demographic factors, such as age, gender, location, or industry experience [18] . These explanations take into account how a job aligns with the user's demographic profile and provide insights into the relevance of the job to specific demographic characteristics. Demographic explanations help users understand how their personal attributes influence the job recommendations they receive.

e) Knowledge/Utility: Knowledge/utility explanations [19] highlight the rationale behind a recommendation by emphasizing key attributes or characteristics of the recommended job. For instance, an explanation might state, "We recommend Job X because it offers a high salary," focusing on the utility aspect of job compensation.

f) Case-based reasoning: Case-based reasoning explanations link current recommendations to similar past cases. They draw parallels with previous instances to justify the current suggestion, showcasing the relevance of historical preferences and choices.

2) Tagsplanations: Tagsplanations [22], involve providing explanations based on the tags or keywords associated with the recommended job and the user's profile. For example in the paper, the provided explanation illustrates an interface featuring a list of tags, accompanied by their relevance to the movie and the user's preferences for each respective tag.

3) Histogram: This approach employs a histogram with a grouping interface [17] and has been mentioned in Section II. Figure 1 shows an example that displays ratings of similar users' movie ratings. As mentioned previously, we can convert this approach to fit the problem of JRSs.Assuming we have access to data, it might be possible to substitute rating values with different metrics that represent the quality of a recommendation. For example, these metrics could encompass actions such as 1 = Scrolled past the Job Posting, 2 = Clicked on the Job Posting, 3 = Submitted an Application for the Job, 4 = Successfully Hired.

4) Pie Chart: This approach utilizes a pie chart to visually represent the same data as the histogram discussed in the preceding section [17](Figure 2). The conversion strategy applicable in the previous context can also be applied here.

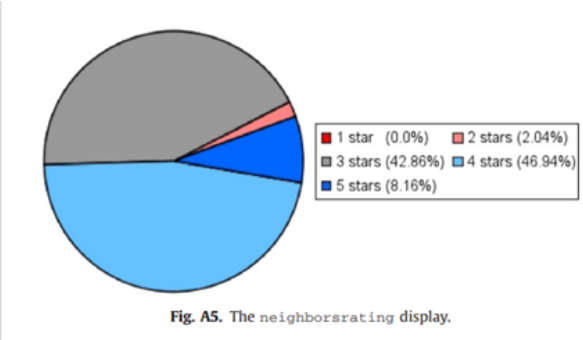


Fig. 2: Piechart with a grouping interface, the categories are a similar user's rating to a film.



Fig. 3: Personalized TagCloud showing different tags and highlighting their importance through color and size [10].

5) Tag Clouds – Personalized: Tag clouds are a visual representation of text data [10], where words or tags are displayed in varying sizes based on their frequency or importance within a dataset(Figure 3). Users can easily identify the key themes or elements driving the recommendation by observing the size and prominence of specific tags within the cloud. Personalized tag clouds are customized for individual users based on their preferences, behavior, or profile data.

6) Tag Clouds – Non-Personalized: Non-personalized tag clouds [10] are generic representations of data or content that do not take individual user characteristics into account. They display tags or keywords that are relevant to the entire dataset or content collection, regardless of individual user preferences.

7) User Rating: User rating explanations are brief descriptions or comments provided by users alongside their numerical ratings for products, services, or content. These explanations offer insights into why a user assigned a particular rating, providing valuable context and information to other potential consumers and helping them make informed decisions.

8) Rating with Confidence Display: Rating with Confidence Display explanations [23] include numerical ratings paired with a confidence measure. These explanations convey not only the user's assessment of the job but also their level of confidence in the rating. Higher confidence suggests a stronger belief in the accuracy of the rating, while lower confidence may indicate uncertainty or limited information.

9) Keyword Style: Keyword-style explanations [23] involve highlighting specific keywords or phrases associated with a job recommendation to emphasize key attributes, skills, or qualifications. These explanations provide a concise summary

of why a particular job is a suitable match for the user based on the presence of relevant keywords in the job posting or the user's profile.



(a) Candidate-side graph (b) Company-side graph

Fig. 4: Explanation in the type of a Knowledge Graph [8]

10) Knowledge Graph: Knowledge Graph explanations [8] leverage interconnected nodes and relationships within the graph to elucidate why a particular job is recommended(Figure 4). Users can explore the graph to understand how their skills, qualifications, or preferences align with the job's attributes, industry, location, or other relevant factors.

11) Feature Attribution/Importance: Feature importance explanations [8] highlight the specific features or attributes that played the most significant roles in predicting an outcome, such as job recommendations. These explanations reveal which factors were the most influential in the recommendation process and can be displayed in a bar plot for example.

B. Use Case 1: Actual implementation

We want to apply the above-mentioned explanation types on a use case to demonstrate their effect and quality properly. Therefore, we designed two use cases. In the first use case, we want to implement an actual job recommender system using machine learning techniques. Our focus here is not a perfect implementation though, rather a simple one for demonstration purposes of the explanation types. Since job recommender systems can work for multi-stakeholders, we designed two different approaches: Once generating job recommendations for future employees, and once generating employee recommendations for companies. In the following section we specify the details on the approaches.

1) Recommendations for employees: To be able to generate job recommendations for employees through machine learning techniques, one would need data first. We will use data about different jobs. Through some Natural Language Preprocessing (NLP) techniques, we will take the CV of an employee and extract information about the employee. Then, using the job dataset, we will recommend the jobs based on similarity with the CV. This way the top n jobs can be displayed for the employee. After that, we want to see which of the above-mentioned explanation types are applicable and which ones are the best. The data we want to use: Job data

2) Recommendations for companies: Using the same similarity idea, we want to generate recommendations of employees for a company that have similar characteristics like the company itself. For that we would need data about different users/employees and information of the example company. For the latter we can use the same data from Job data. However, finding reasonable data about employees is difficult. We have not find a solution for this problem yet. In the worst case, we would drop the recommendations for this stakeholder, as we still have 3 other scenarios where we can apply the explanation types on.

C. Use Case 2: Mock-Up Recommender System

For use case 2, instead of implementing a job recommender system, we want to design a mock up example of such a system. Compared to the actual implementation, so use case 1, it has the advantage that we can design the recommender system so that is does not have flaws that could hinder us using as much explanation types as possible. Given that our focus is not on implementing the recommender system, but rather exploring explanations for job recommender systems, this use case saves us time and helps us concentrate on the actual task of this project.

1) Recommendations for employees: The mock up recommender system for employees, should have data of jobs, similar to use case 1 and a complete and informative CV of the employee. Looking at the similarity of each job with the CV is here as well an approach that we will try. Another idea is to use Collaborative-based Filtering. This technique finds people with similar interests and generates recommendations on this basis. Furthermore, Context-based recommendations, so, generating recommendations based on the user's past interactions with the website, is worth looking into as well. Specifically, a mix of Context-based recommendations and Knowledge-based Filtering can help finding similar jobs based on user's past interactions. The extra knowledge here would be which jobs are similar to each other. Since we have the flexibility to generate both the data and recommendations, we can design a highly customizable and diverse recommender system, which then allows us applying diverse explanations.

2) Recommendations for companies: The mock up recommender system for companies, should have data of jobs seekers, specifically, information from the CV of the job seekers. Furthermore, it is important to not only consider the skills and background of the employees, but also their interests. Data about employees from similar companies can help creating a preferred employee profile and then search for employee that most likely have the same profile. Hence, Collaborative-based Filtering is here again an approach worth exploring.

D. Experiments Set-Up

The first experiment will analyse which explanation types can be applied to explain recommendations of a job recommender system implemented in machine learning. Therefore, we will first create recommendations for this experiment, and try to apply as many explanations as possible. The evaluation of the explanations should focus on correctness, consistency and comprehensibility. The first two can be analysed through manual inspection, however, for the comprehensibility we were thinking of creating a questionnaire and collect as much feedback as possible. If time allows, A/B testing will also be analysed. In other words, experiments where some users receive recommendations with explanations, while others do

not and assess whether or not explanations lead to better decision-making.

The second experiment will analyse which explanation types can be applied to explain recommendations of a mock-up job recommender system. The procedure will be the same as for the first experiment.

IV. Deliverables

In order to record and present our progress and results, there are on the one hand the deliverables, which are mandatory in every project. The blog post in phase 2 and the final project report fall under this category. As the specifications are already set in the assignments those do not need to be defined further in this scope. On the other hand, there are the following project-specific key delivarables:

- **Introduction Session:** As described in use case 1 we will implement a job recommender system, that is able to create suitable job recommendations for employees and in the best case is also able to do employee recommendations for companies. This System will be used to generate recommendations for a dataset which we then use to try out different explanation types that explain how and why those recommendations were generated.
- **Mockup recommender system:** This mock up of a recommender system which we described in use case 2 serves as an example that creates perfect recommendations for both employees and companies and thus allows us to focus on exploring the different explanation types.
- **Test-results:** Creating recommendations with our actual implementation of a recommender system will result in test-results which can be adequately presented. This also counts for the explanations that we will create for these test results. Ideally, our experiments show how to appropriately use explanations for explaining different recommendations by providing clear guidelines and considerations.
- **Scientific Conclusions:** We aim to acquire new insights into the realm of explainable AI within the context of multi-stakeholder job recommender systems, through the scientific collection and presentation of evaluations for both approaches.

V. Time Management and Planning

This section will give a high-level description of the project plan for its three phases. Planning helps in tracking progress in the project which will reduce the chance of falling into delays. The main planning tool used in this project is a Gantt chart divided in the 3 Phases of the Semester:

A. Phase 1

The main idea behind the first phase is researching the core concepts of the project and the approaches throughout the project. The first phase starts by researching the core concepts of the project which are: Explainable Artificial Intelligence(XAI) and Recommendation Systems. After understanding the core concepts, the group will conduct an in-depth research on different types of explanations. Parallel to the explanation research, the group will also research possible data collection methods for both the recommendation system and the experiment. At the end of the phase, the group is expected to provide a project plan and present the findings of the research phase. Figure 5 shows the visualized plan of the first phase.

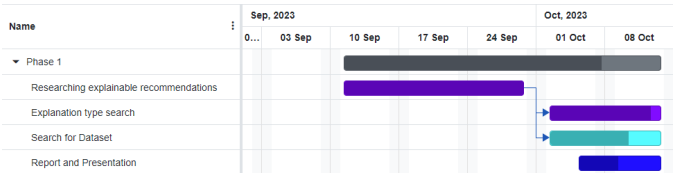


Fig. 5: Gantt chart of the project's first phase.

B. Phase 2

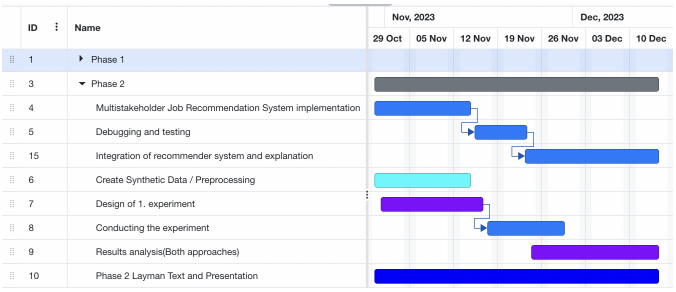


Fig. 6: Gantt chart of the project's second phase.

Phase 2 contains most of the practical work. The goal of this phase is to make all the necessary preparations for the experiments as well as to conduct and start to analyse them. Therefore one part of the group will begin the preparations for the second experiment by implementing the recommendations system. In the case that the search for an actual dataset was not successful, there is also the task of generating synthetic data for this use case. During this time the other part of the group will design and conduct the first experiment. In this phase the results of both approaches will first be analysed individually. Once the implementation of the recommendations system is complete, the group will integrate the explanations from the experiment with the implemented system.

The group will also independently work on the layman text and the presentation which have to be submitted at the end of phase 2.

C. Phase 3

In the third phase, the group will be mainly focusing on evaluating and comparing all the results from the two approaches in phase 2. A part of the group will be finalising the integration of the system and the explanations, while the other part will focus on evaluation. In a first step, the approaches will be evaluated thoroughly one by one. Additionally, both approaches will be compared with each other as far as it is possible. Finally, every group member is going to contribute to finalising the project report and the last presentation.

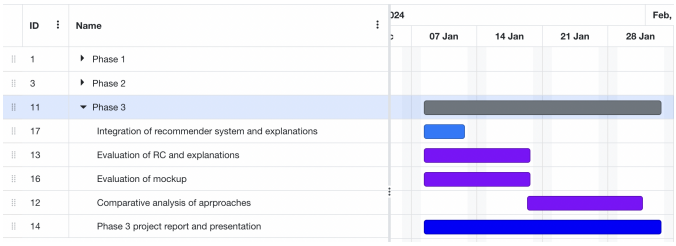


Fig. 7: Gantt chart of the project's third phase.

VI. Risk Management

A. Data

The challenge of insufficient data for multi-stakeholder job recommender systems poses a significant risk to the successful execution of our first use case approach. As highlighted in [8], the dataset utilized was in Mandarin, potentially limiting its applicability for our research within the given time, which may require data in different languages or with different characteristics.

To mitigate this risk, we propose two scenarios:

1) Mitigation Scenario 1: Utilizing the Existing Data: In this scenario, if we find useful data or believe that the existing dataset from the Mandarin source can be adapted for our purposes, we will proceed with using that data to build a recommender system.

However, this approach may introduce concerns about accuracy and performance, given the potential linguistic and cultural differences between the Mandarin dataset and the target audience. It is worth noting that this approach has been implemented in previous work [8] and can serve as a starting point.

2) Mitigation Scenario 2: Generating Synthetic Data: If we determine that the available data is unsuitable for our research objective within the given time, we may generate synthetic applicant and vacancy profiles. This approach would provide us with greater control over the data, allowing us to assume specific distributions or correlations that align with our research goals.

However, the drawback of this approach is that the generated data may not accurately represent real-world job recommendations, potentially impacting the practical applicability of our findings.

In either mitigation scenario, data preprocessing will be necessary. This could involve transforming the Mandarin data into a usable format or mining/scraping sample data that better suits our research needs. Importantly, our primary focus is on improving the explainability of job recommendations, and while the recommendations generated from the adapted or synthetic data may not precisely mirror reality, this discrepancy may be acceptable within the context of our research objectives.

Ultimately, the choice between these mitigation scenarios will depend on the availability and suitability of data and the trade-offs between data accuracy and research focus. It is crucial to carefully consider these options to ensure that our research effectively addresses the gap between XAI and multi-stakeholder job recommender systems.

B. Suitability of Evaluation Methods

The choice between the two distinct use-cases, one involving the development of an actual recommender system and the other focusing on a fully mocked system, poses a risk related to the differences in their nature, which may render direct comparisons challenging. Revolving around the potential inability to compare these two use-cases equally due to their inherit dissimilarity in complexity, the number of explanations, or design.

For example, if one case allows for only a limited number of explanations, while the other permits a greater variety, it can be challenging to devise a fair evaluation strategy. Similarly, if an explanation type can be applied in both use cases, but is kept deliberately simple in one while the other is fully elaborated, the fairness of the comparison may be compromised or void.

1) Mitigation Scenario: To mitigate this risk, we will:

1) **Acknowledge the Inherent Differences**: Clearly acknowledge and document the fundamental differences between the two use-cases, emphasizing that a direct one-to-one comparison may not (always) be feasible or fair.
2) **Perform Within-Use-Case Analysis**: emphasize the importance of conducting in-depth analysis within each use-case to provide insights into their respective context, while not directly comparing them with each other.
3) **Utilizing Common Grounds**: Identifying specific aspects or elements that are common to both use-cases, such as shared explanation types or evaluation criteria, and use these as the basis for comparison.
4) **Qualitative Comparisons**: In cases where quantitative comparisons are challenging, employ qualitative assessments to explore the strengths and weaknesses of each use-case independently.

C. MSRS Performance

One of the main risks associated with the first use-case is the performance of the developed recommender system. The development team might face constraints in resources and data quality, which could affect the system's ability to meet the expectations or standards of the industry.

1) Mitigation Scenario: This risk cannot be mitigated given the previously stated constraints. Fortunately, our primary objective is to ensure explainability, which implies that performance is not a critical factor as long as the system can provide any recommendation, regardless of its relevance or quality. This means that in the case performance is indeed unsatisfactory, we can just accept it as it is.

D. Diversity of User Pool

One of the potential risks of conducting a questionnaire for the first experiment is that the user pool may not be diverse enough to represent the target population. This could lead to biased results and limit the generalizability of the findings.

1) Mitigation Scenario: To mitigate this risk, we will:

1) Use multiple sources and methods to recruit participants, such as online platforms, social media, referrals, incentives, etc.
2) Use appropriate questionnaire design and data collection methods that are suitable for our participants and our research objectives. For example, the use of simple and clear language, avoiding leading or loaded questions, providing multiple options or scales for responses, allowing for open-ended feedback, etc.
3) Analyze and interpret our data carefully and critically. We will consider the limitations and assumptions of our methods and results, and use descriptive and inferential statistics to summarize and test our data. We will also report any potential sources of bias or error and how they may affect our conclusions.

VII. Concluding Remarks

Throughout the first phase of the project, multiple literature reviews and case studies relevant to the topic have been analyzed. We've developed a range of explanation types, each employing distinct approaches aimed at transforming them into MSRS explanations. Furthermore, we've conducted research to establish a general framework for our experiments and evaluations, with a primary emphasis on qualitative data analysis. Lastly, we've identified potential risks associated with our work and devised strategies for their effective mitigation.

Appendix

A. Appendix A

B. Appendix B

C. Appendix C

Using the generated synthetic data, an initial plan was to use Collaborative Filtering to implement a Multi-Stakeholder Explainable Job Recommendation System using python. Collaborative Filtering is a technique used in Recommendation Systems that tends to suggest recommendations based on similar users. The initial step would be to represent all applicants in a tabular form with all the possible skills, and use a numerical representation where an applicant who possesses a skill would have their proficiency score, and 0 otherwise. Similarly, jobs would be represented as rows with the required proficiency level, and 0 if the skill is not included. Using this representation we could find the most suitable applicant for each vacancy and vice-versa.

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